Requirements

Cohort 3
Group 11
Team Aubergine
Joshua Wainwright, Piotr Koziol, Sarvesh Sridhar, Harrison Barrans,
Daniel Thwaites, Callum Newton, Arnav Jamidar, Harry Turner

Introduction

This document describes the project requirements of our team's 2D adventure game, "Escape from Uni". In eliciting these requirements, we met with the client to understand their needs and expectations for the project, and discussed in depth our prepared list of questions. Our general approach mimics the standards and procedures recommended in IEEE 29148-2011, an international standard used in the lifecycle of a programme for requirements engineering.

User requirements were documented from the user-needs highlighted in our discussions with the client. Using non-technical language, they describe the key tasks players should be able to perform. System requirements were derived from User requirements and define how the game will meet these user-needs, given the practical, technical, and creative constraints. Lastly, constraint requirements were scoped from the client's briefing document, and from conclusions drawn about the development environment.

These requirements are recorded in a tabular format for clarity and consistency, supported with a citable ID, and prioritised with the shall/should/may framework to manage scope and feasibility.

This framework was introduced in IEEE 830, a standard for software requirement specification. It provides a way to document the necessity of a requirement. Requirements labelled with ‘Shall’ describe fundamental characteristics which accept and process inputs, and generate outputs; requirements labelled with ‘Should’ are characteristics which would be favourable to have and are recommended to include; requirements labelled ‘May’ could be included, but not at the cost of requirements labelled ‘Shall’ or ‘Should’.

These requirements are complete and singular, and should accurately state a single characteristic without the need of additional context.

To improve requirement traceability, all of our requirements are upwards traceable to a specific requirement or single statement of need (SSoN): functional and non-functional requirements are upwards traceable to user requirements via IDs, and user requirements have a SSoN field. All requirements trace to their source.

We gradually refined this document through routine requirement reviews with the client, ensuring that our requirements and understanding were aligned with the client’s expectations.

USER

ID	Single Statement of Need	Priority
UR-01	The game shall be playable on standard desktop PCs across Windows, Linux, and Mac.	Shall
UR-02	The game shall not require specialist hardware; keyboard and mouse input only.	Shall
UR-03	The game shall be family-friendly, engaging to teenagers and young adults, designed to be simple and accessible for a broad audience.	Shall
UR-04	The game shall run locally and offline, without the need for internet or an account.	Shall
UR-05	The game shall include a short play session of 5 minutes, suitable for casual players with hidden content or events to encourage replayability.	Shall
UR-06	The player shall have a clear means of understanding the impact of interactions (feedback or UI indication).	Shall
UR-07	The game should include a tutorial or introduction (demo, screenshots, or manual)	Shall
UR-08	The game should include SFX or music to enhance immersion.	May
UR-09	The player should be able to pause the game at any time during the base difficulty.	Should
UR-10	The game should include accessibility considerations.	Should
UR-11	The game may include multiple difficulty levels in future versions.	May
UR-12	The website shall host internal project materials and need not be publicly accessible.	Should
UR-13	The game shall include consistent art, theme, and mood suited to its setting.	Shall
UR-14	The game should allow for expansion (i.e., new maps or modes in future iterations).	Should
UR-15	The game should include a leaderboard with top players and their respective highest scores.	Should

UR-16	The game should have achievements that can be achieved upon the completion of a set of in-game tasks	Should

FUNCTIONAL

ID	Description	User
FR-01	The system shall launch and run on standard desktop operating systems using Java to ensure cross-platform compatibility without additional setup.	UR-01
FR-03	The system shall provide menus to launch and close the game, with clear success or failure messages.	UR-09, UR-06
FR-04	The system shall broadcast an internal timer of 5 minutes to the user.	UR-05
FR-05	The system shall provide feedback visually or textually after each user interaction (i.e., item collection, obstacles)	UR-06
FR-06	The system shall include a pause/resume button during gameplay.	UR-09
FR-07	The tutorial screen or demo shall include a sequence explaining controls and objectives.	UR-07
FR-08	The system shall provide a consistent art and UI theme that matches the story and lore.	UR-13
FR-09	The system should include simple background music or SFX for events.	UR-08
FR-10	The system may randomly spawn hidden or humorous events to encourage exploration.	UR-05
FR-11	The system shall be isolated and independent, storing no player data.	UR-04
FR-12	The system should provide accessibility options through built-in design (contrasting colours, readable fonts, audio/text cues).	UR-03
FR-13	The system shall be easily extensible to support new levels or difficulty modes.	UR-14,UR-15, UR-16
FR-14	The website shall host documentation, progress updates, and design artefacts, without complex styling.	UR-12

NON-FUNCTIONAL

ID	Description	User	Fit Criteria
NFR-01	Security: No online features nor data storage, no user accounts.	UR-04	No network connections established; no personal data collected.
NFR-02	Reliability: Game should not crash during playthroughs.	UR-04	Games should remain stable for at least 5 consecutive minutes of playtesting.
NFR-03	Performance: Game should maintain smooth responsiveness.	UR-04	Frame rate should be at least 30 FPS on average desktop computers.
NFR-04	Maintainability: Code shall be well-documented for future developers and extensions.	UR-14	Future developers should be able to add new levels within 2 days of onboarding.
NFR-05	Operability: Game shall be intuitive for new users with clear feedback.	UR-06, UR-07	Player understands basic controls after interacting with the tutorial, and can complete playthroughs unaided.
NFR-06	Accessibility: Game shall support colour-blind and text-readable design.	UR-05	All cues provided through at least 3 sensory modes (VFX, text, audio).
NFR-07	Scalability: Game architecture shall allow additional levels or features.	UR-14	New map textures can be easily introduced using existing entry-points and functions.
NFR-08	Resilience: Game shall handle unexpected input without crashing.	UR-04	Invalid actions trigger friendly error messages or soft-fail states.
NFR-09	Documentation: Website shall include all deliverables, accessible by client.	UR-12	All files uploaded and viewable without permission errors within 12 hours of submission.

CONSTRAINT

Type	Description
Design	The product is a single-player maze-escape game, set in a university-like environment, written in Java and targeted at the Cohort.
Design	Sessions can be paused at any time, and can last a max of 5 minutes, but faster completion increases base score.
Design	Full build includes at least 5 visible hindering, 3 beneficial, and 3 hidden events; all of which may influence the score.
Process	A single event of each type must be functional in time for assessment 1.
Process	ope, time and cost together under the triple-contraint model. If a constraint changes, adjust the rest accordingly.
Process	Maintain accessible project documentation and decision logs for the customer stakeholder throughout development.
Financial	The project must remain within available student time and resources, prioritising core gameplay and accessibility over advanced features.